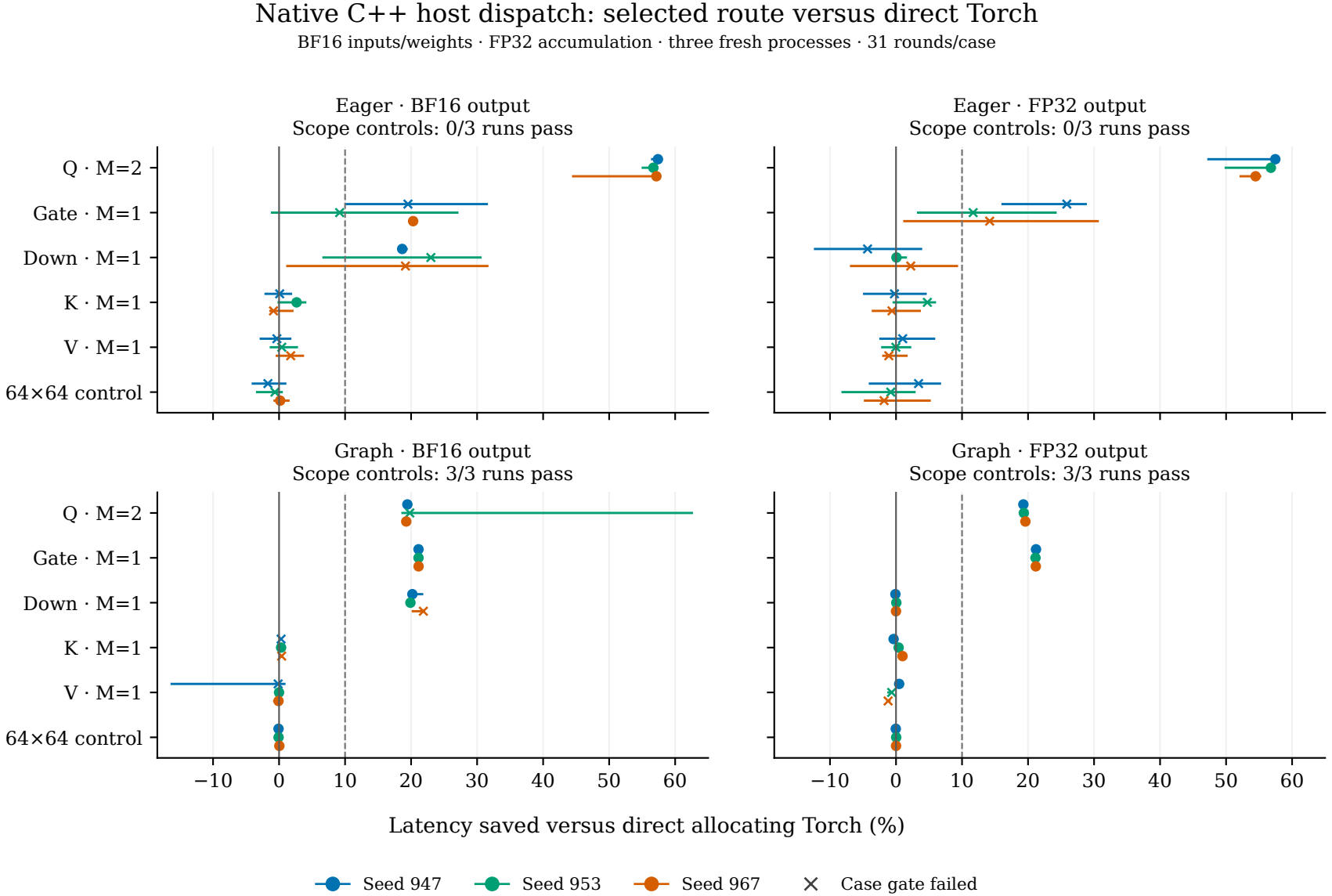
Native C++ host dispatch: selected route versus direct Torch
BF16 inputs/weights · FP32 accumulation · three fresh processes · 31 rounds/case
Eager · BF16 output
Scope controls: 0/3 runs pass
Eager · FP32 output
Scope controls: 0/3 runs pass
Graph · BF16 output
Scope controls: 3/3 runs pass
Graph · FP32 output
Scope controls: 3/3 runs pass
Q · M=2
Gate · M=1
Down · M=1
K · M=1
V · M=1
64×64 control
−10
0
10
20
30
40
50
60
Latency saved versus direct allocating Torch (%)
Seed 947
Seed 953
Seed 967
Case gate failed
Bars: paired 95% intervals. Eager includes allocations/sync; graph replay excludes host and allocation cost.
Stock RTX 3070 Ti Laptop · Performance on AC · memory 810 MHz · failed controls prevent general API acceptance